Automating Irrigation Distribution Uniformity Assessments in California Almonds Using UAVs: Land Surface Temperature Corrections

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Poster References, Appendix, & Contact Info

Introduction

Motivation

- California is home to 80% of the world's almond production [1]
- Almonds require 101-156 cm/yr of water, but receives 25-75 cm/yr of precipitation [2]
- Irrigation Distribution Uniformity (DU) – evenness of water distribution across a field; time-taxing; poor DU wastes water and nutrients

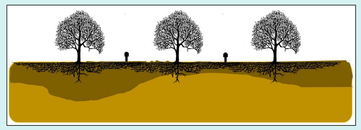


Figure 1. Poor Irrigation DU Illustration: low irrigation DU can easily cause nonuniformity in yield and plant water stress [3]

- Under-irrigation (and other factors) cause plant water stress, land surface temperature (LST) increases

So, if they're related, can we map DU using drones, too?

- Drones can map LST, but Forward Looking Infrared (FLIR) cameras have accuracy challenges; Micasense Altum PT Thermal and Multispectral camera has accuracy of ±5˚C, which needs calibration for DU assessment
- Long flights (30-40 min) approaching solar noon in arid & semiarid climates lead temperature gradients in radiometric temperature (LST)
- Need to correct for this time difference for more accurate interpretation of results
- Goal: create a more representative spatial input of LST

Methodology

Original LST Raster

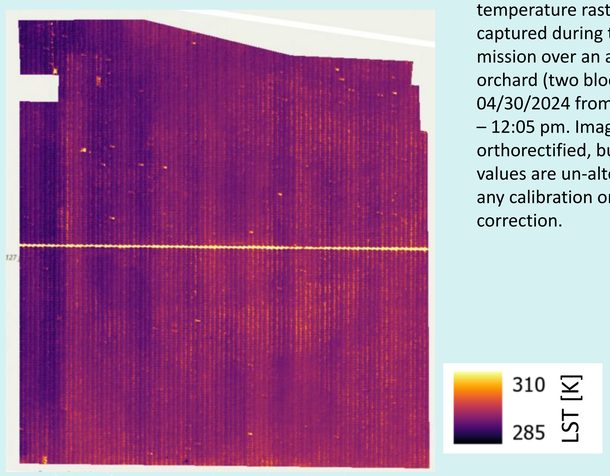


Figure 2. Land surface temperature raster captured during the UAV mission over an almond orchard (two blocks) on 04/30/2024 from 11:36 am – 12:05 pm. Imagery is orthorectified, but pixel values are un-altered for any calibration or thermal correction.

- Large (62 ha), mature almond orchard
- Middle bright band is a dirt road
- Large orchards: LST can change over flight time (start Northwest corner, ends on East side)
- Temperatures rises leading up to solar noon (~1 pm PST)
- UAV system: Micasense Altum PT thermal and multispectral camera mounted on a DJI Matrice 300 RTK

① Extract UAV-measured LST values of calibration tarps on ground from thermal raster and collect IRT data points from same locations (ground truth)

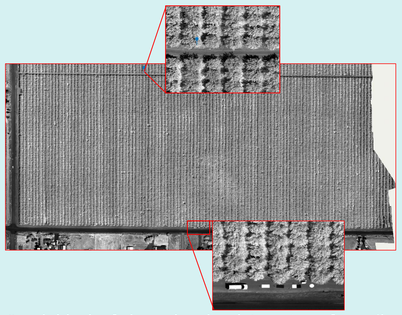


Figure 3. South block of almond orchard. Location of tarp (bottom) and canopy (top) IRT setup. Includes the white tarp, patch of bare soil unobstructed by shadow during UAV mission, and black tarp each with IRTs at 45-deg angle. All tarps are calibrated to have certain reflectance.



Figure 4. Photo of Apogee Infrared Radiometer (IRT) setup for all ground-truth points.

Infrared thermometers estimate the temperature of a surface and capture measurements that are minimally altered by the atmosphere [4]



② Create calibration curve between UAV- and ground-based thermal measurements and calibrate original LST pixel values using this curve

Calibration Regression Curve: Linear, Quadratic, and Logarithmic Fit Results (Excluding Black Tarp Data)

Model	Equation	R-Squared	RSE	p-value
Linear	$y = 92.76 + 0.71x$	0.6005	4.325	0.00186
Quadratic	$y = -5688 + 37.76x - 0.0593x^2$	0.8073	3.150	0.00027
Logarithmic	$y = -967.7 + 223.3*\log(x)$	0.6085	4.282	0.00166

Sensor-Corrected LST Raster

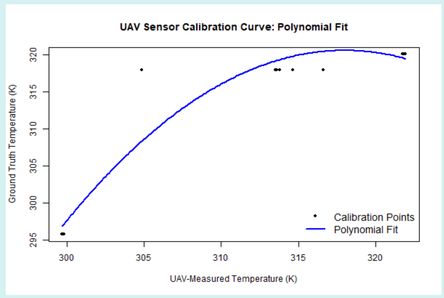


Figure 5. Second-degree polynomial regression between white tarp, canopy, and bare soil IRT data (ground truth) and UAV-measured data. Equation was applied to the original LST raster to create the sensor-corrected LST raster (left).

③ Build a time series of pixel values from UAV flight according to time captured based on flight path position

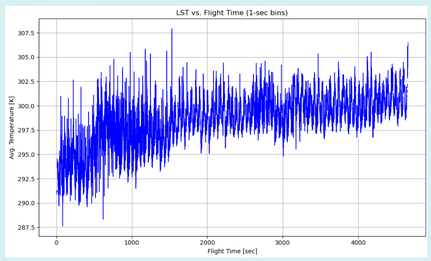


Figure 6. Time series of sensor-corrected LST raster.

④ Build combined linear equation for IRT data over flight time to establish time-temperature relationship

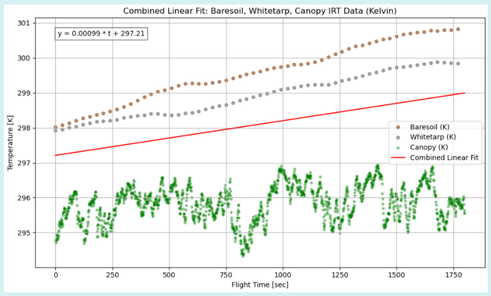


Figure 7. Linear fit curve for time calibration using IRT data over flight time.

Results

⑤ Rebuild the raster using the time-corrected pixel values

Sensor- & Time-Corrected LST Raster

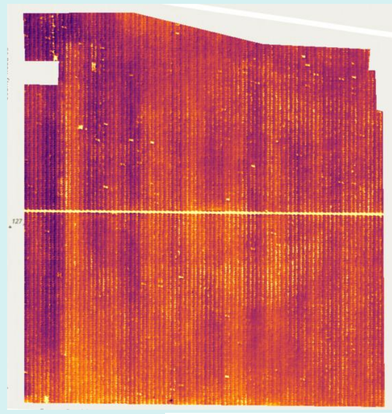


Figure 8. Sensor- and time-corrected LST raster using IRT data.

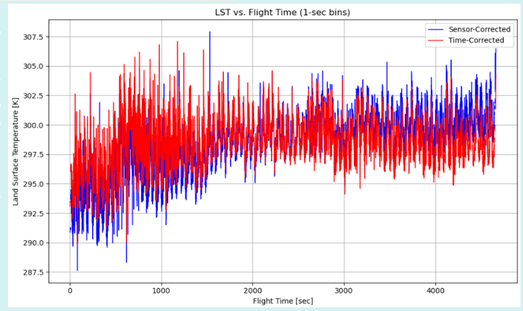


Figure 9. Time series before and after the time-correction. Blue is before (only sensor-corrected), and red is after (sensor- and time-corrected).

Conclusions

- This protocol corrected for FLIR camera accuracy issues and reduced the gradient effects of the canopy warming up during the UAV mission duration
- The combined linear fit for time correction is weakened by leaf-level temperature fluctuations

Future Direction

- Validate this protocol with new imagery at various almond orchards in the California Central Valley collected June 2025
- Refine protocol to use for all radiometric temperature inputs in the TSEB model [5] to more precisely and accurately assess the spatial variability of evapotranspiration
- Use these results to build a python-based model to automate DU assessments for CA almonds

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